

13. PHOTOSYNTHESIS IN HIGHER PLANTS

EXPERIMENTS RELATED WITH PHOTOSYNTHESIS

NEET: Previous Years Questions

- Oxygen is not produced during photosynthesis by
 (1) Green-sulphur bacteria (2) *Nostoc*
 (3) *Cycas* (4) *Chara* (**NEET 2018**)

Model Questions

- Photosynthesis is
 (1) A chemical process by which green plants use light energy to synthesise organic compounds.
 (2) A physico-chemical process by which green plants use light energy to synthesise organic compounds.
 (3) A chemical process by which green plants use light energy to synthesise inorganic compounds.
 (4) A physico-chemical process by which green plants use light energy to synthesise inorganic compounds.
- Variegated leaf experiment using starch shows that
 (1) Photosynthesis occurs only in green parts of the leaves in presence of light.
 (2) CO_2 is required for photosynthesis.
 (3) Glucose is produced when plants grow and it is usually stored as starch.
 (4) Glucose is made in the green parts of plants.
- CO_2 is required for photosynthesis. The chemical used to remove CO_2 from entering a control apparatus is
 (1) Calcium oxide (2) Distilled water
 (3) KOH solution (4) Na_2CO_3
- Which of the following with respect to early experiments of photosynthesis is wrongly matched?
 (1) Priestley – Showed that plants release oxygen.
 (2) Jan Ingenhousz – Showed that sunlight is essential for photosynthesis.
 (3) Julius von Sachs – Proved that plants produce glucose when they grow.
 (4) T.W. Engelmann – Showed that the green substance is located within special bodies in plants.
 (5) Cornelius van Niel – Showed that hydrogen reduces CO_2 to carbohydrates.
- In an experiment demonstrating the evolution of oxygen in *Hydrilla*, sodium bicarbonate is added to the water in the experimental setup. What would happen if all other conditions are favourable?
 (1) Amount of oxygen evolved decreases as the availability of CO_2 increases.
 (2) Amount of oxygen evolved increases as CO_2 in water is absorbed by sodium bicarbonate.
 (3) Amount of oxygen evolved decreases as CO_2 in water is absorbed by sodium bicarbonate.
 (4) Amount of oxygen evolved increases as the availability of CO_2 increases.
- In Purple & green sulphur bacteria, oxygen is not evolved during photosynthesis because hydrogen donor is
 (1) H_2O (2) H_2S (3) NH_3 (4) CH_4

PHOTOSYNTHESIS: SITE AND PIGMENTS

NEET: Previous Years Questions

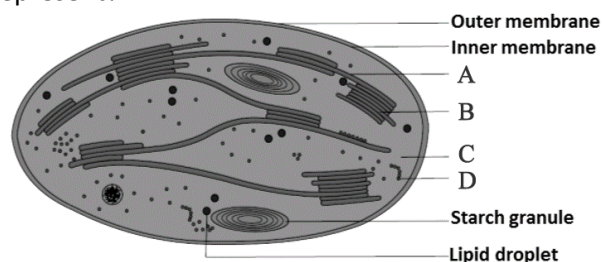
- The reaction centre PS II has an absorption maxima at
 (1) 700 nm (2) 660 nm
 (3) 780 nm (4) 680 nm (**NEET 2023**)
- Emerson's enhancement effect and Red drop have been instrumental in the discovery of (**NEET-I 2016**)
 (1) photophosphorylation and cyclic electron transport
 (2) oxidative phosphorylation
 (3) photophosphorylation and non-cyclic electron transport
 (4) two photosystems operating simultaneously.

Model Questions

- Assertion:** Chloroplasts are present in the walls of mesophyll cells of leaves.

Reason: It helps to get optimum quantity of incident light.

- Both assertion and reason are true and reason is the correct explanation of assertion.
 - Both assertion and reason are true but reason is not the correct explanation of assertion.
 - Assertion is true but reason is false.
 - Both assertion and reason are false.
- In the given figure of chloroplast, what do A, B, C & D represent?



- (1) A- Stromal lamella, B- Stroma, C- Grana, D- Ribosomes
 (2) A- Grana, B- Stromal lamella, C- Stroma, D- Ribosomes
 (3) A- Stromal lamella, B- Grana, C- Stroma, D- Ribosomes
 (4) A- Stroma, B- Grana, C- Stromal lamella, D- Ribosomes
12. Chlorophyll molecules are located in the
 (1) Thylakoid lumen
 (2) Thylakoid membrane
 (3) Stroma
 (4) Inner chloroplast membrane
13. Match the following:
 A. Chlorophyll a (i) Yellow
 B. Chlorophyll b (ii) Bright or blue green
 C. Xanthophyll (iii) Yellow- Yellow orange
 D. Carotenoids (iv) Yellow green
 (1) A- (ii), B- (iv), C- (i), D- (iii)
 (2) A- (iii), B- (iv), C- (ii), D- (i)
- (3) A- (iv), B- (iii), C- (ii), D- (i)
 (4) A- (iv), B- (ii), C- (i), D- (iii)
14. The absorption spectrum & action spectrum coincide closely showing that photosynthesis is maximum at
 (1) Blue & green regions of the spectrum.
 (2) Green & red regions of the spectrum.
 (3) Blue & red regions of the spectrum.
 (4) Blue & violet regions of the spectrum.
15. The chief pigment associated with photosynthesis is
 (1) Chlorophyll *a* (2) Chlorophyll *b*
 (3) Xanthophyll (4) Carotenoids
16. Regarding photosystems, pick out the wrong statement.
 (1) Each photosystem has a chlorophyll *a* and accessory pigments bound by proteins.
 (2) All pigments, except one molecule of chlorophyll *a*, form a light harvesting complex.
 (3) Single chlorophyll *a* acts as reaction centre.
 (4) In PS I, the reaction centre absorbs light at 680 nm, and so called P680.

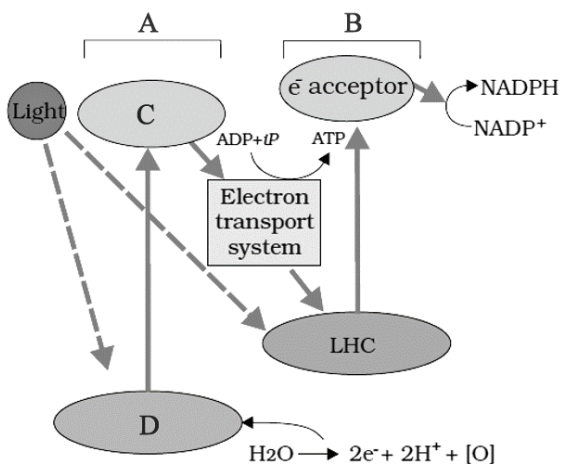
LIGHT REACTION (PHOTOCHEMICAL PHASE)

NEET: Previous Years Questions

17. Which of the following combinations is required for chemiosmosis?
 (1) Membrane, proton pump, proton gradient, NADP synthase
 (2) Proton pump, electron gradient, ATP synthase
 (3) Proton pump, electron gradient, NADP synthase
 (4) Membrane, proton pump, proton gradient, ATP synthase
(NEET 2023)
18. Which of the following statements is incorrect?
 (1) Both ATP and NADPH + H^+ are synthesized during noncyclic photophosphorylation.
 (2) Stroma lamellae have PS I only and lack NADP reductase.
 (3) Grana lamellae have both PS I and PS II.
 (4) Cyclic photophosphorylation involves both PS I and PS II.
(NEET 2021)
19. In light reaction, plastoquinone facilitates the transfer of electrons from:
 (1) PS-I to NADP+
 (2) PS-I to ATP synthase
 (3) PS-II to Cytb6f complex
 (4) Cytb6f complex to PS-I
(NEET 2020)
20. Which of the following is not a product of light reaction in photosynthesis?
 (1) ATP (2) NADH
 (3) NADPH (4) Oxygen
(NEET 2018)
21. In a chloroplast the highest number of protons are found in
 (1) intermembrane space
 (2) antennae complex
 (3) stroma
 (4) lumen of thylakoids
(NEET-I 2016)

Model Questions

22. **Assertion:** In light reaction of photosynthesis, light is required for the functioning of PS I and PS II, and production of NADPH and ATP.
Reason: Dark reaction does not occur in light.
 (1) Both assertion and reason are true and reason is the correct explanation of assertion.
 (2) Both assertion and reason are true but reason is not the correct explanation of assertion.
 (3) Assertion is true but reason is false.
 (4) Both assertion and reason are false.
23. Light reactions do not include
 (1) Water splitting
 (2) Use of NADPH to synthesise food
 (3) Oxygen release
 (4) Formation of ATP & NADPH
24. Transfer of electrons from PS II to PS I and finally downhill to $NADP^+$ is called
 (1) Z scheme (2) C4 pathway
 (3) C3 pathway (4) Photorespiration
25. Which of the following is correctly labelled for the given figure?



- (1) A- LHC, B- $\bar{e}$ acceptor, C- PS I, D- PS II
 (2) A- PS II, B- PS I, C- $\bar{e}$ acceptor, D- LHC
 (3) A- PS I, B- PS II, C- $\bar{e}$ acceptor, D- LHC
 (4) A- $\bar{e}$ acceptor, B- LHC, C- PS II, D- PS I
26. Pick out the wrong statement.
- (1) The water splitting complex in PS II is located on the inner side of the thylakoid membrane.
 (2) The protons formed by water splitting are needed to replace the protons that are moved from PS II.
 (3) The protons (H^+) are used to reduce NADP to NADPH.
 (4) The PS II provides electrons needed to replace those removed from PS I.
27. Which of the following is an essential factor for photophosphorylation?
- (1) Sunlight (2) Carbohydrate
 (3) Oxygen (4) Water
28. Non-cyclic photo-phosphorylation results in the production of
- (1) ATP (2) ADP & NADPH + H^+
 (3) ATP & NADPH + H^+ (4) NADPH
29. Incorrect statement regarding Cyclic photo-phosphorylation is
- (1) It occurs in stroma lamellae when only PS I is functional.
 (2) Here, ATP and NADPH + H^+ is synthesised.
 (3) The stroma lamellae membranes lack PS II and NADP reductase.
 (4) The electron does not pass on to NADP $^+$ but is cycled back to PS I complex through electron transport chain.
30. In chloroplast, highest number of protons are found in
- (1) Intermembrane space
 (2) Antenna complex
 (3) Stroma
 (4) Lumen of thylakoids

31. According to the chemiosmotic hypothesis, ATP production is the result of
- (1) Proton gradient developed across the membrane of thylakoids.
 (2) Electron gradient developed across the membrane of thylakoids.
 (3) Proton gradient developed in stroma.
 (4) Electron gradient developed in stroma.
32. Splitting of water occurs on the
- (1) Outer side of the thylakoid membrane
 (2) Inner side of the thylakoid membrane.
 (3) Stroma of the chloroplast
 (4) Cytoplasm
33. Which of the following is not required for chemiosmosis?
- (1) A membrane
 (2) A proton pump & A proton gradient
 (3) Bundle sheath cells
 (4) ATP synthase
34. During the light-dependent reactions of photosynthesis, the primary electron acceptor associated with photosystem II (PSII) transfers electrons to an H carrier. This process directly results in:
- (1) Release of protons into the stroma.
 (2) Uptake of protons from the lumen.
 (3) Removal of protons from the stroma and release of protons into the lumen.
 (4) Movement of electrons directly to NADP $^+$.
35. The proton gradient across the thylakoid membrane is crucial for ATP synthesis. Which of the following statements accurately describes the establishment and utilization of this gradient?
- (1) Protons accumulate in the stroma, leading to a decrease in stromal pH and driving ATP synthesis through F1.
 (2) The breakdown of the proton gradient involves the movement of protons from the stroma to the lumen through ATP synthase.
 (3) The proton gradient is established by the pumping of protons into the lumen, leading to a decrease in lumen pH, and its breakdown powers ATP synthesis via proton movement through F0 of ATP synthase to the stroma.
 (4) The primary electron acceptor pumps protons directly into the stroma, creating the gradient used by ATP synthase.
36. Which of the following correctly describes the roles of the F₀ and F₁ parts of ATP synthase during oxidative phosphorylation or photosynthesis?

- (1) F_0 – ATP synthesis; F_1 – Proton channel
 (2) F_0 – Conformational change to release ATP; F_1 – Proton gradient breakdown

- (3) F_0 – Facilitated diffusion of protons; F_1 – ATP synthesis
 (4) F_0 – Binds ADP and inorganic phosphate; F_1 – Proton pumping.

DARK REACTION (BIOSYNTHETIC PHASE) - USE OF ATP & NADPH

NEET: Previous Years Questions

37. Which of the following are required for the dark reaction of photosynthesis? **(NEET 2024)**

- A. Light B. Chlorophyll C. CO_2
 D. ATP E. NADPH

Choose the correct answer from the options given below:

- (1) A, B and C only
 (2) B, C and D only
 (3) C, D and E only
 (4) D and E only

38. How many molecules of ATP and NADPH are required for every molecule of CO_2 fixed in the Calvin cycle?

- (1) 2 molecules of ATP and 3 molecules of NADPH
 (2) 2 molecules of ATP and 2 molecules of NADPH
 (3) 3 molecules of ATP and 3 molecules of NADPH
 (4) 3 molecules of ATP and 2 molecules of NADPH

(NEET 2024)

39. Given below are two statements:

(NEET 2024)

Statement I: In C_3 plants, some O_2 binds to *RuBisCO*, hence CO_2 fixation is decreased.

Statement II: In C_4 plants, mesophyll cells show very little photorespiration while bundle sheath cells do not show photorespiration.

In the light of the above statements, choose the correct answer from the options given below:

- (1) Both Statement I and Statement II are true
 (2) Both Statement I and Statement II are false
 (3) Statement I is true but Statement II is false
 (4) Statement I is false but Statement II is true

40. How many ATP and $NADPH_2$ are required for the synthesis of one molecule of Glucose during Calvin cycle?

- (1) 18 ATP and 12 $NADPH_2$
 (2) 12 ATP and 16 $NADPH_2$
 (3) 18 ATP and 16 $NADPH_2$
 (4) 12 ATP and 12 $NADPH_2$

(NEET 2023)

41. The first stable product of CO_2 fixation in *sorghum* is:

- (1) Pyruvic acid (2) Oxaloacetic acid
 (3) Succinic acid (4) Phosphoglyceric acid

(NEET 2021)

42. The oxygenation activity of *RuBisCO* enzyme in photorespiration leads to the formation of:

- (1) 1 molecule of 6-C compound
 (2) 1 molecule of 4-C compound and 1 molecule of 2-C compound
 (3) 2 molecules of 3-C compound
 (4) 1 molecule of 3-C compound **(NEET 2020)**

43. Phosphoenol pyruvate (PEP) is the primary CO_2 acceptor in **(NEET 2017)**

- (1) C_4 plants (2) C_2 plants
 (3) C_3 and C_4 plants (4) C_3 plants

44. The process which makes major difference between C_3 and C_4 plants is **(NEET-II 2016)**

- (1) glycolysis (2) Calvin cycle
 (3) photorespiration (4) respiration

45. A plant in your garden avoids photorespiratory losses, has improved water use efficiency, shows high rates of photosynthesis at high temperatures and has improved efficiency of nitrogen utilisation. In which of the following physiological groups would you assign this plant?

- (1) CAM (2) Nitrogen fixer
 (3) C_3 (4) C_4 **(NEET-I 2016)**

Model Questions

46. Which of the following is not a product of light reaction?

- (1) ATP (2) NADPH
 (3) O_2 (4) CO_2

47. Select the wrong statement regarding dark reaction.

- (1) It is the use of ATP and NADPH to drive the processes for the synthesis of food.
 (2) This phase directly depends on the light.
 (3) It is dependent on the products of the light reaction.
 (4) It is also known as Biosynthetic phase.

48. Two types of CO_2 assimilation during photosynthesis are

- (1) C_3 pathway and C_4 pathway
 (2) Light reaction and dark reaction
 (3) Anabolic pathway and catabolic pathway
 (4) Cyclic and Non-cyclic phosphorylation

49. First stable product of CO_2 fixation in C_3 pathway is

- (1) 3-phosphoglyceric acid (PGA)
 (2) Oxaloacetic acid (OAA)
 (3) Phosphoenol pyruvate (PEP)

- (4) Ribulose-1,5-bisphosphate (RuBP)
50. C_3 pathway was discovered by
- (1) Louis Pasteur (2) Melvin Calvin
 - (3) Hatch and Slack (4) Robert Hill
51. First stable product in C_4 pathway is
- (1) Ribulose-1,5-bisphosphate (RuBP)
 - (2) Phosphoenol pyruvate (PEP)
 - (3) 3-phosphoglyceric acid (PGA)
 - (4) Oxaloacetic acid (OAA)
52. Pick out the false statement about C_3 pathway.
- (1) It occurs only in C_3 plants.
 - (2) It includes 3 stages such as carboxylation, reduction & regeneration.
 - (3) It does not have Kranz anatomy.
 - (4) In this, RuBP is the primary CO_2 acceptor.
53. Most crucial step of C_3 pathway is
- (1) Carboxylation of RuBP
 - (2) Reduction
 - (3) Regeneration of RuBP
 - (4) Photorespiration
54. The enzyme RuBP carboxylase is also termed "RuBisCO" because it:
- (1) Catalyzes only the carboxylation of ribulose-1,5-bisphosphate (RuBP) in the Calvin cycle.
 - (2) Exhibits dual activity, functioning as both a carboxylase and an oxygenase.
 - (3) Is involved in the regeneration of RuBP during the Calvin cycle.
 - (4) Facilitates CO_2 fixation exclusively in C_4 plants.
55. During the reduction phase of the Calvin cycle, how many ATP and NADPH molecules are required to synthesize one glucose molecule?
- (1) 12 ATP & 12 NADPH (2) 18 ATP & 12 NADPH
 - (3) 12 ATP & 18 NADPH (4) 6 ATP & 6 NADPH
56. In the Calvin cycle, the regeneration of ribulose-1,5-bisphosphate (RuBP) requires ATP. This difference in the stoichiometry of ATP and NADPH consumption during the dark reaction is compensated by which of the following processes?
- (1) Non-cyclic photophosphorylation
 - (2) Photorespiration
 - (3) Cyclic photophosphorylation
 - (4) Oxidative phosphorylation
57. To make 1 glucose molecule, how many turns of the Calvin cycle are needed?
- (1) 1 (2) 2 (4) 4 (4) 6
58. In the Calvin cycle, which of the following correctly represents the inputs and outputs for the synthesis of one molecule of glucose?
- (1) **Inputs:** 6 CO_2 , 18 ATP, 12 NADPH;
- Outputs:** 1 glucose, 18 ADP, 12 $NADP^+$
- (2) **Inputs:** 6 CO_2 , 12 ATP, 18 NADPH;
 - Outputs:** 1 glucose, 12 ADP, 18 $NADP^+$
 - (3) **Inputs:** 12 CO_2 , 18 ATP, 12 NADPH;
 - Outputs:** 2 glucose, 18 ADP, 12 $NADP^+$
 - (4) **Inputs:** 3 CO_2 , 9 ATP, 6 NADPH;
 - Outputs:** 1 glucose, 9 ADP, 6 $NADP^+$
59. Kranz anatomy in C_4 plants is characterized by:
- (1) Thin-walled mesophyll cells with abundant chloroplasts
 - (2) Bundle sheath cells arranged in a ring around vascular tissues
 - (3) Presence of air spaces in bundle sheath cells for gas exchange
 - (4) Vascular bundles surrounded by palisade parenchyma
60. In C_4 plants, the main biosynthetic pathway responsible for carbohydrate synthesis is:
- (1) C_3 pathway (Calvin cycle)
 - (2) C_4 pathway (Hatch & Slack pathway)
 - (3) CAM pathway
 - (4) Glycolytic pathway
61. In the Hatch and Slack (C_4) pathway, the primary CO_2 acceptor and the enzyme responsible for its fixation in mesophyll cells are:
- (1) Ribulose-1,5-bisphosphate (RuBP) and *RuBisCO*
 - (2) Phosphoenol pyruvate (PEP) and *PEP carboxylase*
 - (3) Oxaloacetic acid (OAA) and *Malate dehydrogenase*
 - (4) 3-phosphoglyceric acid (PGA) and *Pyruvate kinase*
62. Why do mesophyll cells in C_4 plants lack the Calvin cycle?
- (1) They do not contain the enzyme RuBisCO.
 - (2) They lack NADPH required for the Calvin cycle.
 - (3) They are specialized only for ATP synthesis.
 - (4) They cannot transport malic acid to bundle sheath cells.
63. In C_4 plants, bundle sheath cells play a critical role in photosynthesis. Which of the following statements is correct regarding these cells?
- (1) Bundle sheath cells contain PEP carboxylase (PEPcase) and are the primary site for CO_2 fixation into C_4 acids.
 - (2) Bundle sheath cells lack RuBisCO but use the C_4 pathway to generate glucose directly.
 - (3) Bundle sheath cells are rich in RuBisCO and facilitate the Calvin cycle, while mesophyll cells regenerate PEP.
 - (4) Bundle sheath cells transport C_3 acids to mesophyll cells for the synthesis of oxaloacetic acid (OAA).

64. Which of the following is a **correct** feature of C_4 plants?
- (1) They exhibit high photorespiration and low temperature tolerance.
 - (2) They lack Kranz anatomy but have high biomass productivity.
 - (3) They perform photosynthesis only under low light intensities.
 - (4) They possess Kranz anatomy and lack photorespiration, leading to higher biomass production.
65. In Calvin pathway, the reaction catalyzed by Rubisco is correctly represented by
- (1) $\text{RuBP} + \text{CO}_2 \rightarrow 3 \times 3\text{PGA}$
 - (2) $\text{RuBP} + \text{O}_2 \rightarrow 2 \times 3\text{PGA}$
 - (3) $\text{RuBP} + \text{CO}_2 \rightarrow 2 \times 3\text{PGA}$
 - (4) $\text{RuBP} + \text{O}_2 \rightarrow 3 \times 3\text{PGA}$
66. The most abundant enzyme in the world is
- (1) RuBisCO
 - (2) Catalase
 - (3) PEP Carboxylase
 - (4) ATP Synthase
67. Photorespiration occurs when RuBisCO binds to O_2 instead of CO_2 . Which factor primarily determines whether *RuBisCO* acts as a *carboxylase* or an *oxygenase*?
- (1) The pH of the chloroplast stroma
 - (2) The relative concentrations of CO_2 and O_2
 - (3) The availability of ATP and NADPH
 - (4) The enzyme's higher affinity for O_2
68. In C_3 plants, photorespiration occurs because:
- (1) RuBisCO has a higher affinity for O_2 than CO_2
 - (2) RuBP binds with O_2 , forming phosphoglycerate and phosphoglycolate
 - (3) RuBisCO fixes CO_2 at a higher rate in the presence of O_2
 - (4) RuBP directly binds to CO_2 , reducing O_2 fixation
69. Photorespiration is a wasteful process because
- (1) It synthesizes ATP and NADPH without producing sugars.
 - (2) It uses ATP and releases CO_2 without synthesizing sugars, ATP, or NADPH.
 - (3) It prevents CO_2 fixation by inhibiting RuBisCO activity.
 - (4) It consumes ATP and NADPH to synthesize sugars inefficiently.
70. Photorespiration does not occur in C_4 plants because
- (1) RuBisCO in C_4 plants lacks *oxygenase* activity.
 - (2) CO_2 is directly fixed into sugars in the mesophyll cells.
 - (3) C_4 plants maintain a high CO_2 concentration at RuBisCO's enzyme site.
 - (4) Oxygen concentration is reduced in the mesophyll cells.
71. What advantages do C_4 plants have over C_3 plants?
- (1) Higher RuBisCO activity, higher oxygenase activity and better water conservation
 - (2) Faster growth due to higher oxygenase activity of RuBisCO
 - (3) Better CO_2 fixation at low temperatures and presence of photorespiration
 - (4) Increased productivity, tolerance to higher temperatures, and absence of photorespiration
72. Which of the following statements correctly distinguishes C_3 plants from C_4 plants?
- (1) C_3 plants have dimorphic chloroplasts, while C_4 plants have only granal chloroplasts.
 - (2) C_4 plants have a higher CO_2 compensation point compared to C_3 plants.
 - (3) Photosynthesis in C_4 plants occurs in both mesophyll and bundle-sheath cells, while in C_3 plants, it occurs only in mesophyll cells.
 - (4) C_3 plants are more photosynthetically efficient and have higher productivity than C_4 plants.
73. In C_3 plants and C_4 plants, optimum temperature for photosynthesis is
- | | |
|---|---|
| (1) C_3 plants: $30^\circ\text{C} - 40^\circ\text{C}$ | C_4 plants: $35^\circ\text{C} - 45^\circ\text{C}$ |
| (2) C_3 plants: $35^\circ\text{C} - 45^\circ\text{C}$ | C_4 plants: $30^\circ\text{C} - 40^\circ\text{C}$ |
| (3) C_3 plants: $35^\circ\text{C} - 45^\circ\text{C}$ | C_4 plants: 25°C |
| (4) C_3 plants: 25°C | C_4 plants: $35^\circ\text{C} - 45^\circ\text{C}$ |
74. Which of the following sets are completely C_4 plants?
- (1) Maize, rice, wheat, sorghum
 - (2) Maize, sugarcane, amaranth, sorghum
 - (3) Rice, wheat, bean, potato
 - (4) Rice, bean, potato, sugarcane

FACTORS AFFECTING PHOTOSYNTHESIS

NEET: Previous Years Questions

75. With reference to factors affecting the rate of photosynthesis, which of the following statements is not correct?
- (1) Increasing atmospheric CO_2 concentration up to 0.05% can enhance CO_2 fixation rate.
 - (2) C_3 plants respond to higher temperature with enhanced photosynthesis while C_4 plants have much lower temperature optimum.
 - (3) Tomato is a greenhouse crop which can be grown in CO_2 enriched atmosphere for higher yield.
 - (4) Light saturation for CO_2 fixation occurs at 10% of full sunlight.

(NEET 2017)

Model Questions

76. Which of the following combinations correctly categorizes internal (plant) and external factors affecting photosynthesis?
- (1) **Internal:** Number of leaves, chlorophyll amount
External: Temperature, soil pH
 - (2) **Internal:** Mesophyll cells, internal CO_2 concentration
External: Sunlight, atmospheric CO_2
 - (3) **Internal:** Leaf orientation, water availability
External: Genes, growth rate
 - (4) **Internal:** Age of leaves, soil nutrients
External: Light intensity, humidity
77. "If a biochemical process is affected by more than one factor, its rate is determined by the factor nearest to its minimal value: it is the factor which directly affects the process if its quantity is changed." This is called
- (1) Liebig's Law of the Minimum
 - (2) Blackman's Law of Limiting Factors
 - (3) Law of Segregation
 - (4) Shelford's Law of Tolerance
78. Which of the following statements best describes the relationship between light intensity and the rate of CO_2 fixation in photosynthesis?
- (1) CO_2 fixation rate increases linearly with light intensity indefinitely, even at very high light levels.
 - (2) Light saturation occurs at 10% of full sunlight, after which CO_2 fixation plateaus due to limiting factors like temperature or CO_2 availability.
 - (3) At high light intensities, chlorophyll synthesis increases, leading to a proportional rise in photosynthesis.
 - (4) Light is the primary limiting factor for photosynthesis in all plants, regardless of their habitat.
79. Regarding carbon dioxide concentration, which statement is wrong?
- (1) CO_2 is the major limiting factor for photosynthesis.
 - (2) Increase in CO_2 concentration in atmosphere beyond 0.05% can become damaging for plants.
 - (3) At low light, C_3 and C_4 plants respond to high CO_2 .
 - (4) C_3 plants respond to increased CO_2 concentration and saturation is seen only beyond $450 \mu\text{L}^{-1}$.
80. Why are the dark reactions of photosynthesis more sensitive to temperature changes compared to the light reactions?
- (1) Chlorophyll stability decreases at higher temperatures.
 - (2) Light reactions depend on enzymatic processes.
 - (3) Electron transport in light reactions is inhibited by heat.
 - (4) Dark reactions involve temperature-sensitive enzymatic steps.
81. Which of the following correctly describes the temperature adaptation of C_3 and C_4 plants?
- (1) C_3 plants have a higher temperature optimum than C_4 plants.
 - (2) C_4 plants perform better at lower temperatures due to PEP carboxylase.
 - (3) C_4 plants thrive at higher temperatures, while C_3 plants have lower optima.
 - (4) Both C_3 and C_4 plants have the same temperature optimum globally.
82. Which of the following is the primary reason for reduced photosynthesis during water stress in plants?
- (1) Increased stomatal aperture enhances CO_2 loss.
 - (2) Enhanced chlorophyll synthesis in wilted leaves.
 - (3) Reduced CO_2 availability and decreased leaf surface area.
 - (4) Accelerated photorespiration due to high O_2 concentration.

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